

[T]-Theory: A Universal Theory of Everything

One Equation. 61 Decades. Mind, Body, and Cosmos.

Why: neural activity is not only a collection of neurons firing; it is also a field that integrates activity across tissue and time. [T]-Theory treats the cortical electromagnetic field as a testable bridge between local signals, global coordination, and felt state.

Proved (Lean 4, 0 sorries):

- SHO = Green's function (derived, not postulated as a primitive)
- 11D manifold = minimum geometry for a conscious vertebrate organism
- Scale invariance $10^{-35}\text{m} \rightarrow 10^{26}\text{m}$ (61 orders of magnitude)
- Theory is a proposed fixed-point attractor of its own subject matter (see the [T]-Theory phenomena paper)

Numerical hits (Planck 2018):

- $\Omega_\Lambda = 7/11$ (93.2%) — dark energy as compactified vacuum
- $\Omega_c = 3/11$ (97.1%) — dark matter as spatial vacuum; gravity only; no direct detection possible
- $w = -1$ exact — live knife-edge test: DESI DR1 & Euclid

[T]-Theory accounts for 95.5% of the total mass-energy of the universe via a zero-parameter geometric derivation.

I • The Master Field Equation

The Universal Somatic Field is governed by a single structural equation — the Green's function of a tensor field derived from M-theory compactification.

Stationary form (Helmholtz):

$$(\nabla^2 + k^2) G(x, x') = -\delta^3(x - x')$$

Time-dependent form (d'Alembertian):

$$\left(\frac{1}{v_s^2} \partial_t^2 - \nabla^2 + k^2 \right) \Phi_{\mu\nu}(x, t) = -J_{\mu\nu}(x, t)$$

Retarded propagator — causal, $\tau = t - t' > 0$:

$$G_R(r, \tau) = \frac{v_s}{4\pi r} e^{-kv_s\tau} \delta(\tau - r/v_s) \theta(\tau)$$

Somatic Memory Kernel:

$$K(\tau) = K_0 e^{-\tau/\tau_m} \theta(\tau), \quad \tau_m = \frac{1}{kv_s}$$

General solution (field = integral of all past sources):

$$\Phi(x, t) = \Phi^{(0)}(x, t) + \int_{-\infty}^t dt' \int d^3x' G_R J$$

The **effective mass** k sets the spatial correlation length $\ell = 1/k$. As $k \rightarrow 0$ (near consciousness threshold T_c), $\ell \rightarrow \infty$: global integration onset.

The **field temperature** T_{field} controls how freely the field explores the attractor landscape. Therapeutic window: $T_{\text{field}} \in [T_{\text{min}}, T_{\text{max}}]$.

II • Type-Theoretic Architecture (HoTT / Lean 4)

The soma-field has a rigorous type-theoretic formulation in Homotopy Type Theory, machine-verified in Lean 4 using Mathlib and Physlib.

The Σ -type — soma-field as a fiber bundle over the 20-scale base:

$$\text{SomaField} \equiv \sum_{(\sigma : \text{Scale}_{20})} \text{Substrate}(\sigma)$$

where $\text{Substrate}(\sigma) : \text{Type}$ is the physical substrate at scale σ . The base space is the 20-point scale hierarchy; each fiber is a field configuration.

The Zoom Operator — dependent type constructor between adjacent fibers:

$$\Lambda : (\sigma : \text{Scale}_{20}) \rightarrow \text{Substrate}(\sigma) \rightarrow \text{Substrate}(\sigma + 1)$$

Causality as a proof argument — the retarded propagator's full type:

$$G_R : (t \ t' : \text{Time}) \rightarrow (t' < t) \rightarrow \text{Space} \rightarrow \text{Space} \rightarrow \text{Field}$$

The inequality $t' < t$ is a *proof term* — the Lean 4 kernel enforces the arrow of time at compile time. A causal violation is a type error, not a runtime error.

The full USF type:

$$\text{USF} \equiv \sum_{(\sigma : \text{Scale}_{20})} (\text{Substrate}(\sigma) \times G_R(\sigma))$$

BFSS cortex — coordinates emerge from Hermitian